Aiden Zhou

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EDUCATION

Yale University Expected: **May 2028**
Bachelor of Science in **Computer Science** and **Mathematics** GPA: **3.93/4.0**

- **Relevant Coursework:** Algorithms, Data Structures, Probability Theory, Advanced Linear Algebra, Real Analysis, Discrete Math, Computational Vision, Graph Deep Learning, Fourier Analysis
- **Leadership:** Multimedia Managing Editor of the *Yale Daily News*, Features Editor of the *Yale Scientific Magazine*, Teaching Assistant for Calculus I/II and Real Analysis

EXPERIENCE

Cohere | Machine Learning Intern **May 2026 – Present**

- Machine Learning Intern on Cohere's Sovereign AI team, contributing to foundational model post-training and evaluation for enterprise and government use cases

Supervised Program for Alignment Research (SPAR) | Research Intern **Feb 2026 – May 2026**

- Developed adaptive benchmarking methods for LLM evaluation, reducing evaluation cost by up to 98% and variance by 50%–70% relative to random sampling; paper under review at NeurIPS 2026
- Integrated and deployed three Pyro / py-irt response models for ordinal and continuous scoring tasks, expanding adaptive evaluation across agentic and safety benchmarks

Wu Tsai Institute at Yale University | Research Intern **Jan 2025 – Oct 2025**

- Engineered and tested inverse graphics vision models using PyTorch, Nvdiffrast, and MATLAB
- Built an end-to-end ETL pipeline to clean and regularize 50+ GB of ECoG data from 60+ patients across 85 tasks; ran CUDA/Slurm GPU workflows and cut manual workload by ~5x

Algoverse AI | Machine Learning Researcher **May 2025 – Aug 2025**

- Authored and presented a paper at MechInterp @ NeurIPS 2025 on persistent trends in learning dynamics and feature evolution across 143 checkpoints of Pythia LLMs

PROJECTS

Graded LLM Evaluation with Item Response Theory (github.com/aidenzhou8/Ordinal-FBM)

- Extended Fluid Benchmarking beyond binary outcomes to support ordinal and continuous IRT methods such as GRM and GPCM, enabling efficient evaluation for a broader class of benchmarks

Adaptive Interaction Graphs for Particle Simulation (github.com/aidenzhou8/AdaptGNS)

- Developed AdaptGNS, a particle simulator that expands its interaction graph based on local uncertainty, improving long-horizon fluid and granular simulation; paper under review at AI4Physics @ ICML 2026

NeuronEvolution: Tracking LLM Features over Pre-Training (neuronevolution.streamlit.app)

- Deployed a web app to track feature evolution across 143 checkpoints of Pythia LLMs; implemented Python-based backend and used Snowflake and AWS to ingest model activations into a data warehouse

SKILLS

- **Languages:** Python, C/C++, Java, JavaScript, MATLAB, SQL, HTML/CSS, Bash, Racket
- **Machine Learning:** PyTorch, scikit-learn, PyG, Transformers, CUDA, vLLM, Pyro, py-irt, RLVR
- **Systems / Infrastructure:** AWS, GCP, Git, Docker, Kubernetes, Weights & Biases, Grafana, Slurm

HONORS AND AWARDS

- **Yale First-Year Summer Research Scholarship in the Sciences and Engineering (2025)**
- **Math and Science Olympiads:** AIME (3x), CMO, UBC Physics Gold (2x)
- **Chess:** US National Master; 4th at the 2022 North American Youth Championship, 34th at the 2022 World Youth Championship, and top-30 finishes at the World Open and North American Open